

IT Employee Management System

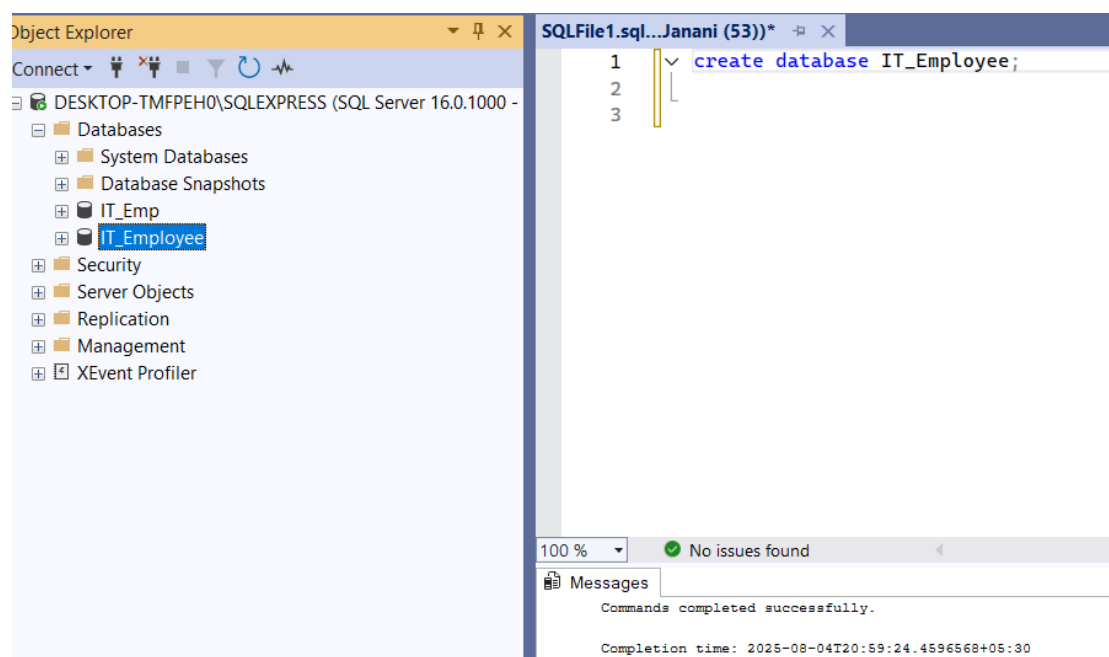
Step 1: Database Schema Requirements

Entities:

1. **Employee**
2. **Department**
3. **Project**
4. **Skill**
5. **EmployeeSkill** (Many-to-Many)
6. **ProjectAssignment** (Many-to-Many)

Step 2: SQL Script to Create the Schema

Creating Database – IT_Employee



Department Table:

The screenshot displays the SQL Server Enterprise Manager interface on the left, showing the 'Databases' folder expanded to 'IT_Employee', and the 'Tables' folder expanded to 'dbo.Department'. The right pane shows a SQL Query window with the following code:

```
1  --create database IT_Employee;
2
3  use IT_Employee;
4  CREATE TABLE Department (
5      DepartmentID INT PRIMARY KEY IDENTITY(1,1),
6      DepartmentName VARCHAR(100) NOT NULL
7  );
8
9
```

The status bar at the bottom indicates 'No issues found' and 'Commands completed successfully.'

Employee Table:

The screenshot displays the SQL Server Enterprise Manager interface on the left, showing the 'Databases' folder expanded to 'IT_Employee', and the 'Tables' folder expanded to 'dbo.Employee'. The right pane shows a SQL Query window with the following code:

```
1  use IT_Employee;
2
3  CREATE TABLE Employee (
4      EmployeeID INT PRIMARY KEY IDENTITY(1,1),
5      FirstName VARCHAR(50),
6      LastName VARCHAR(50),
7      Email VARCHAR(100) UNIQUE,
8      Phone VARCHAR(15),
9      HireDate DATE,
10     JobTitle VARCHAR(100),
11     DepartmentID INT,
12     FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID)
13 );
```

The status bar at the bottom indicates 'No issues found' and 'Commands completed successfully.' The completion time is shown as '2025-08-04T21:12:05.5095921+05:30'.

Project Table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'DESKTOP-TMFPEH0\SQLEXPRESS (SQL Server 16.0.1000 -)'. The 'Tables' folder under 'IT_Employee' is expanded, showing 'dbo.Project' as the selected table. The main pane displays the SQL script for creating the 'Project' table:

```
1 CREATE TABLE Project (  
2     ProjectID INT PRIMARY KEY IDENTITY(1,1),  
3     ProjectName VARCHAR(100),  
4     StartDate DATE,  
5     EndDate DATE  
6 );
```

Below the script, the 'Messages' pane shows the execution results:

```
100 % No issues found  
Messages  
Commands completed successfully.  
Completion time: 2025-08-04T21:12:45.7491853+05:30
```

Skill Table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'DESKTOP-TMFPEH0\SQLEXPRESS (SQL Server 16.0.1000 -)'. The 'Tables' folder under 'IT_Employee' is expanded, showing 'dbo.Skill' as the selected table. The main pane displays the SQL script for creating the 'Skill' table:

```
1 CREATE TABLE Skill (  
2     SkillID INT PRIMARY KEY IDENTITY(1,1),  
3     SkillName VARCHAR(100)  
4 );
```

Below the script, the 'Messages' pane shows the execution results:

```
100 % No issues found  
Messages  
Commands completed successfully.  
Completion time: 2025-08-04T21:16:41.1221878+05:30
```

Employee Skill Table:

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the Object Explorer shows the database structure for 'DESKTOP-TMFPEH0\SQLEXPRESS (SQL Server 16.0.1000 -)'. The 'Tables' folder under 'dbo' is expanded, and 'dbo.EmployeeSkill' is highlighted. The main pane shows the SQL query editor with the following code:

```
1 CREATE TABLE EmployeeSkill (  
2     EmployeeID INT,  
3     SkillID INT,  
4     ProficiencyLevel VARCHAR(20),  
5     PRIMARY KEY (EmployeeID, SkillID),  
6     FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID),  
7     FOREIGN KEY (SkillID) REFERENCES Skill(SkillID)  
8 );
```

The Messages pane at the bottom indicates that the command was completed successfully.

Completion time: 2025-08-04T21:18:56.4889453+05:30

Project Assignment Table:

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the Object Explorer shows the database structure for 'DESKTOP-TMFPEH0\SQLEXPRESS (SQL Server 16.0.1000 -)'. The 'Tables' folder under 'dbo' is expanded, and 'dbo.ProjectAssignment' is highlighted. The main pane shows the SQL query editor with the following code:

```
1 CREATE TABLE ProjectAssignment (  
2     EmployeeID INT,  
3     ProjectID INT,  
4     Role VARCHAR(100),  
5     AllocationPercentage INT CHECK (AllocationPercentage BETWEEN 0 AND 100),  
6     PRIMARY KEY (EmployeeID, ProjectID),  
7     FOREIGN KEY (EmployeeID) REFERENCES Employee(EmployeeID),  
8     FOREIGN KEY (ProjectID) REFERENCES Project(ProjectID)  
9 );
```

The Messages pane at the bottom indicates that the command was completed successfully.

Completion time: 2025-08-04T21:19:27.7899851+05:30

Step 3: ER Diagram Description

Here's how the ER diagram will look:

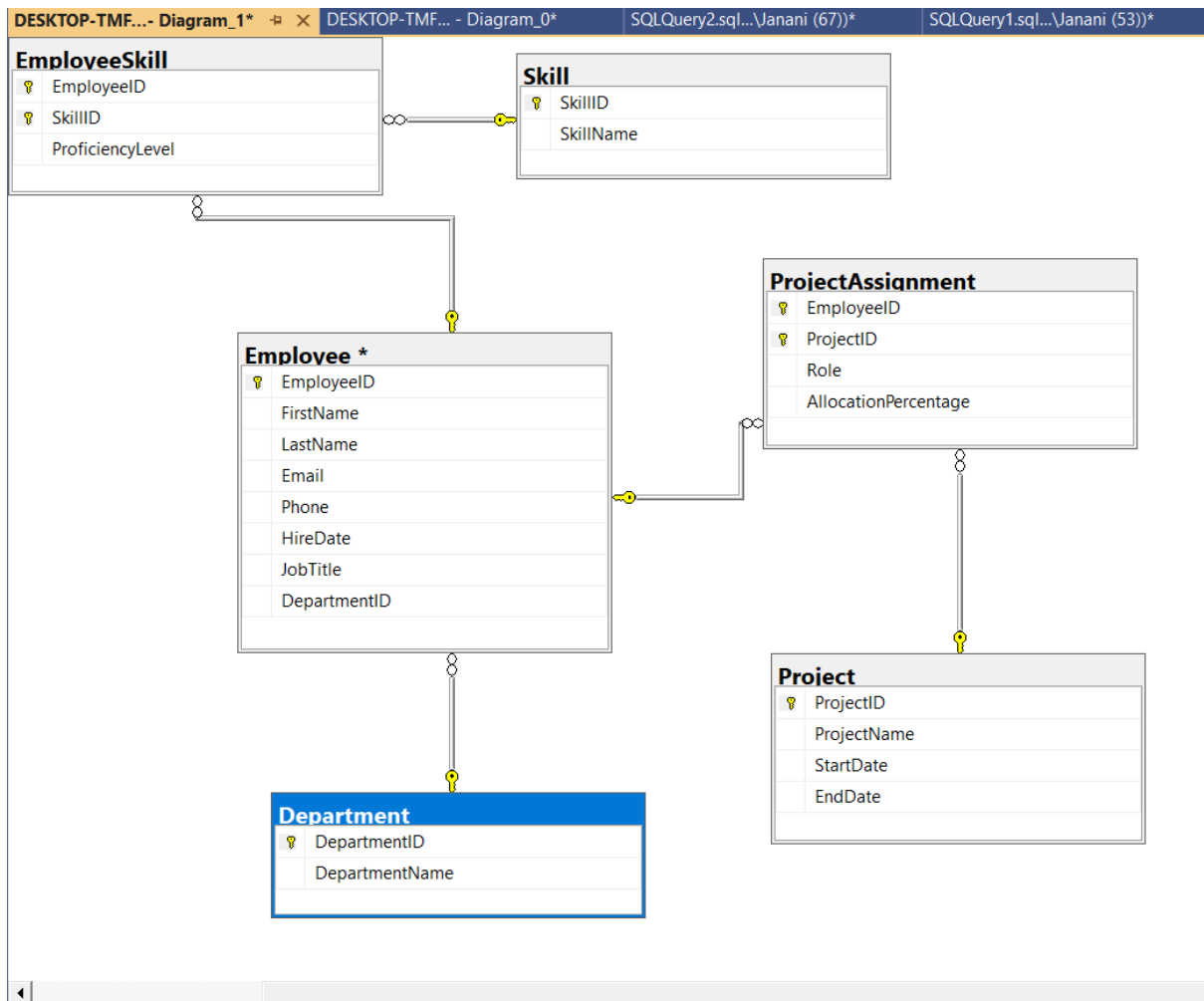
Entities and Attributes

- **Department**
 - DepartmentID (PK)
 - DepartmentName
- **Employee**
 - EmployeeID (PK)
 - FirstName
 - LastName
 - Email
 - Phone
 - HireDate
 - JobTitle
 - DepartmentID (FK)
- **Project**
 - ProjectID (PK)
 - ProjectName
 - StartDate
 - EndDate
- **Skill**
 - SkillID (PK)
 - SkillName
- **EmployeeSkill**
 - EmployeeID (FK)
 - SkillID (FK)
 - ProficiencyLevel

- **ProjectAssignment**
 - EmployeeID (FK)
 - ProjectID (FK)
 - Role
 - AllocationPercentage

Relationships

- **Department** —< **Employee** (One-to-Many)
- **Employee** —< **EmployeeSkill** >— **Skill** (Many-to-Many)
- **Employee** —< **ProjectAssignment** >— **Project** (Many-to-Many)



THANK YOU!